

Ch 7: Linearly Solvable MDPs

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* MDP = Markov Decision Problems

We will formulate it in a slightly more general form than Todorov's original formulation.

What is linearly solvable MDPs?

- It is a special class of **KL control problems** that can be solved efficiently via linear backward dynamic programming.
- It can also be solved by forward Monte Carlo simulations.
- Its generalization to continuous-time, continuous-state setting will be equivalent to path integral control.

Assume

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- discrete time $t = 0, 1, 2, \dots, T$
- finite state space $\mathcal{X} = \{1, 2, \dots, N\}$
- finite control space $\mathcal{U} = \{1, 2, \dots, M\}$

Given :

- $P(x_{t+1} | x_t, u_t) :$
- $Q(u_t | x_t) :$
- $R(u_t | x_t) :$
(also called nominal or behavioral policy)

Joint (trajectory - level) distribution :

$$Q(x_{0:T}, u_{0:T}) = \prod_{t=0}^{T-1} P(x_{t+1} | x_t, u_t) Q(u_t | x_t) P(x_0)$$

— ①

$$R(x_{0:T}, u_{0:T}) = \prod_{t=0}^{T-1} P(x_{t+1} | x_t, u_t) R(u_t | x_t) P(x_0)$$

— ②

Cost function :

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$$\mathbb{E}^Q C(x_t, u_t) = \sum_{x_t, u_t} Q(x_t, u_t) C(x_t, u_t)$$

$$\mathbb{E}^Q D(Q(\cdot | x_t) \| R(\cdot | x_t))$$

$$= \sum_{x_t, u_t} Q(x_t, u_t) \log \frac{Q(u_t | x_t)}{R(u_t | x_t)}$$

KL control problem :

$$\begin{aligned} \min_{Q(u_t | x_t)} \quad & \mathbb{E}^Q \sum_{t=1}^{T-1} \left\{ C(x_t, u_t) + \alpha D(Q(\cdot | x_t) \| R(\cdot | x_t)) \right\} \\ & + \mathbb{E}^Q C_T^{\text{exit}}(x_T) \end{aligned} \quad \text{--- (3)}$$

or equivalently

$$\begin{aligned} \min_Q \quad & \mathbb{E}^Q \left[\sum_{t=1}^{T-1} C(x_t, u_t) + C_T^{\text{exit}}(x_T) \right] \\ & + \alpha D(Q_{x_{0:T}, u_{0:T-1}} \| R_{x_{0:T}, u_{0:T-1}}) \end{aligned} \quad \text{--- (4)}$$

Trajectory-level distributions (1), (2)

Solving KL control by dynamic programming 14

- Define the value function

$$V_t(x_t) :=$$

$$\min_{\{Q(u_k|x_k)\}_{k=t}^{T-1}} \sum_{k=t}^{T-1} \sum_{x_k, u_k} Q(x_k) Q(u_k|x_k) \left\{ c + \alpha \log \frac{Q}{R} \right\}$$

- Value function should satisfy Bellman's equation

$$\begin{aligned} V_k(x_k) &= \min_{Q(x_k|u_k)} \sum_{u_k} Q(u_k|x_k) \left\{ c(x_k, u_k) + \alpha \log \frac{Q(u_k|x_k)}{R(u_k|x_k)} \right. \\ &\quad \left. + \sum_{x_{k+1}} p(x_{k+1}|x_k, u_k) V_{k+1}(x_{k+1}) \right\} \end{aligned}$$

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(7.20)

The optimal policy is computed by

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$$Q^*(u_k | x_k) = \frac{R(u_k | x_k) \exp(-\rho_k(x_k, u_k)/\alpha)}{\sum_{u'_k} R(u'_k | x_k) \exp(-\rho_k(x_k, u'_k)/\alpha)}$$

⇒ Algorithm 11.

Linearly solvable MDPs

The KL control problem becomes linearly solvable under the following additional assumption.

Assumption 7.1 (Deterministic transition)

$$p(x_{t+1} | x_t, u_t) = \begin{cases} 1 & \text{if } x_{t+1} = F(x_t, u_t) \\ 0 & \text{otherwise} \end{cases}$$

$$\rho_k(x_k, u_k) = C(x_k, u_k) + V_{k+1}(F(x_k, u_k))$$

(7.20) becomes

Then

$$\left\{ \begin{array}{l} Z_k(x_k) = \\ \sum_{u_k} R(u_k|x_k) \exp\left(-\frac{C(x_k, u_k)}{\alpha}\right) Z_{k+1}(F(x_k, u_k)) \\ \text{Terminal condition:} \\ Z_T(x_T) = \exp\left(-\frac{C_T^{\text{exit}}(x_T)}{\alpha}\right) \end{array} \right. \quad (7.25)$$

If we write $Z_k = \begin{bmatrix} Z_k(1) \\ \vdots \\ Z_k(N) \end{bmatrix}$

then (7.25) can be written as

$$Z_k = M_k Z_{k+1}$$

for some matrix M_k defined by (7.27).

Hence (7.25) is linearly solvable.

Once Z_k , $k = T, T-1, \dots, 0$ are computed, the optimal policy is computed as

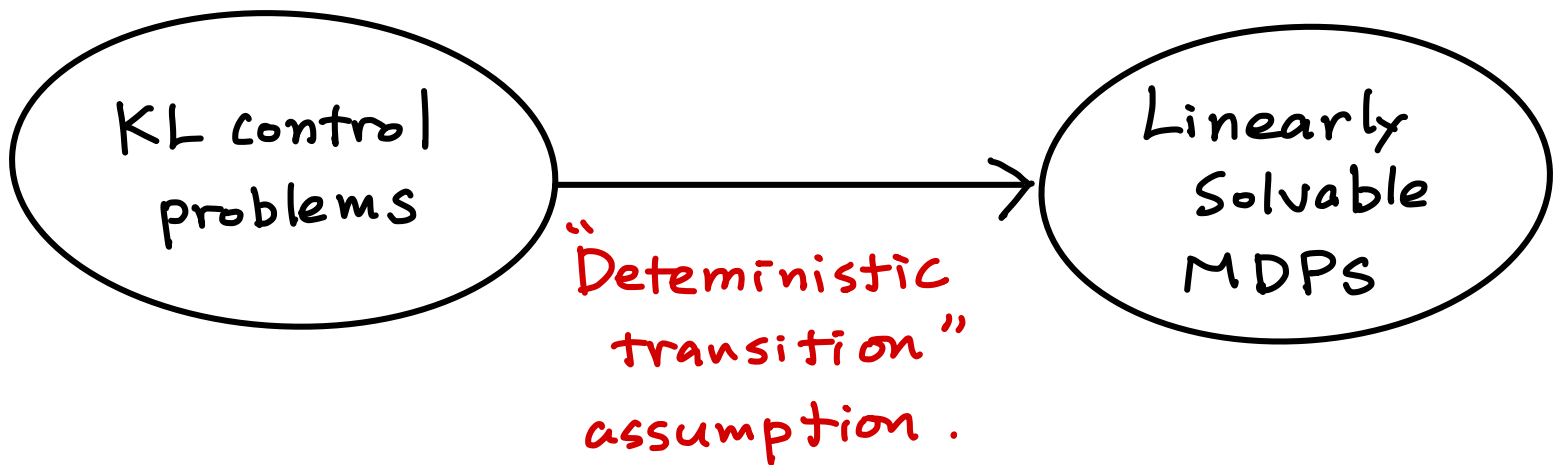
$$Q^*(u_k|x_k) = \frac{R(u_k|x_k) \exp\left(-\frac{C(x_k, u_k)}{\alpha}\right) Z_{k+1}(F(x_k, u_k))}{Z_k(x_k)} \quad (7.29)$$

Algorithm 12 summarizes optimal

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KL control synthesis by linear backward recursion.

We have shown that



Q: What if the state transition kernel was not deterministic?

A: The linear recursion (7.25) only provides an upper bound of the true desirability function. See Exercise 7.3.

$$Z_k(x_k) = \sum_{u_k} R(u_k | x_k) \exp\left(-\frac{C_k(x_k, u_k)}{\alpha}\right) Z_{k+1}(x_{k+1})$$

The original value function is

$$V_k(x_k) = -\alpha \log \left\{ \mathbb{E}^R \exp\left(-\frac{C_{k:T}(x_{k:T}, u_{k:T-1})}{\alpha}\right) \right\}$$

MC algorithm to compute $Z_k(x_k)$:

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Generate N independent sample paths

$$x_k \xrightarrow{u_k} x_{k+1} \xrightarrow{u_{k+1}} x_{k+2} \cdots \xrightarrow{u_{T-1}} x_T$$

all starting from x_k using the reference policy $R(u_t|x_t)$, $t = k, k+1, \dots, T-1$.

Define

$$r(x_k) := \frac{1}{N} \sum_{i=1}^N \exp\left(-\frac{C_{k:T}(x_{k:T}(i), u_{k:T-1}(i))}{\alpha}\right)$$

Then, by the strong LLN,

$$r(x_k) \xrightarrow{\text{a.s.}} Z_k(x_k) \text{ as } N \rightarrow \infty.$$

Computing $Q^*(u_k|x_k)$

Define

$$r(x_k, u_k) := \frac{1}{N} \sum_{i \in I(u_k)} \exp\left(-\frac{C_{k:T}(\quad)}{\alpha}\right)$$

↑

indices of paths in which action u_k is taken.

$$\text{Note: } r(x_k) = \sum_{u_k} r(x_k, u_k)$$

By the strong LLN, it can be shown that 110

$$\frac{r(x_k, u_k)}{r(x_k)} \xrightarrow{\text{a.s.}} Q^*(u_k | x_k)$$

as $N \rightarrow \infty$. (Exercise 7.4)

Algorithm 13 summarizes the optimal KL control synthesis by Monte Carlo simulations.

Q: What is the advantage of Algorithm 13 (forward Monte Carlo) over Algorithm 12 (backward linear recursion)?

A: Algorithm 12 is difficult to implement if $\mathcal{X}$ is continuous.

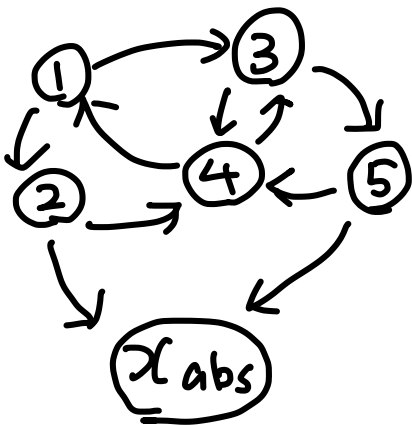
However, Algorithm 13 is easily generalizable to continuous $\mathcal{X}$.

⇒ See Algorithm 14. Further generalization to continuous-time settings will be the path integral control.

Reinforcement learning algorithms

	General case	linearly solvable case
Model-based	value iteration (VI)	
Sim-driven	Monte Carlo RL	
Data-driven	Q-learning	

Setup: Consider a state space with an **absorbing state** x_{abs} .



- $\mathcal{X} = \{1, 2, \dots, N, x_{abs}\}$
 $\mathcal{U} = \{1, 2, \dots, M\}$
- Deterministic transition
$$x_{t+1} = F(x_t, u_t)$$
- Staying at x_{abs} incurs no cost.

Variable end-time problem :

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$$\begin{cases} V^\mu(x) = \mathbb{E}^\mu \left[\sum_{t=0}^{T_f-1} C(X_t, u_t) + \alpha \log \frac{\mu(u_t | x_t)}{\mu'(u_t | x_t)} \mid x_0 = x \right] \\ V^\mu(x_{abs}) = 0 \end{cases}$$

where

- μ' is a time-invariant reference policy
- μ is the policy to be designed
- T_f is the first exit time, i.e.,

$$T_f = \min \{ t \mid x_t = x_{abs} \}$$

Let $V^*(x)$ be the time-invariant value function corresponding to the optimal time-invariant policy.

We know the Bellman equation in terms of

$Z^*(x) = \exp\left(-\frac{V^*(x)}{\alpha}\right)$ is linear.

$$\begin{cases} Z^*(x_t) = \sum_{u_t} \mu'(u_t | x_t) \exp\left(-\frac{C(x_t, u_t)}{\alpha}\right) Z^*(x_{t+1}) \\ Z^*(x_{abs}) = 1 \end{cases} \quad \text{--- (5)}$$

If we write

$$\bar{z}^* = \begin{cases} z^*(1) \\ \vdots \\ z^*(N) \end{cases} \quad \underline{13}$$
$$1 = z^*(x_{abs})$$

⑤ can be written as

$$\underbrace{\begin{bmatrix} z^*(1) \\ \vdots \\ z^*(N) \end{bmatrix}}_{\bar{z}^*} = \underbrace{\begin{bmatrix} a_{11} & \dots & a_{1N} \\ \vdots & & \vdots \\ a_{N1} & \dots & a_{NN} \end{bmatrix}}_A \underbrace{\begin{bmatrix} z^*(1) \\ \vdots \\ z^*(N) \end{bmatrix}}_{\bar{z}^*} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_N \end{bmatrix}}_w$$

where

$$a_{ij} = \sum_{u \in \mathcal{U}_{i \rightarrow j}} \mu'(u|i) \exp\left(-\frac{C(i,u)}{\alpha}\right)$$

$$w_i = \sum_{u \in \mathcal{U}_{i \rightarrow x_{abs}}} \mu'(u|i) \exp\left(-\frac{C(i,u)}{\alpha}\right)$$

Hence, if we have a model information A, w , then solving this optimal control problem reduces to solving a linear system

$$\bar{z}^* = A \bar{z}^* + w$$

Q : What if A and W are unknown,
and the cost $C(x_t, u_t)$ can only be
measured through experiments ?

The following simple algorithm works:

- Keep generating transitions $x_t \rightarrow u_t \rightarrow x_{t+1}$
under the reference policy μ' .
- Update the estimate $\hat{Z}$ of Z^* online as

where θ_t is a diminishing learning rate. — (6)

- (6) is called the **Z-learning**.